



# ESP-Fi HAR: A low-power WiFi CSI dataset for Ad-Hoc IoT human activity recognition<sup>☆</sup>

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## ABSTRACT

Deploying human activity recognition (HAR) systems in large-scale, privacy-preserving ad-hoc IoT networks is constrained by the high cost and power consumption of conventional WiFi sensing hardware. To address this challenge, we present ESP-Fi HAR, a low-power WiFi channel state information (CSI) dataset collected using commercial ESP32 modules. The dataset covers four indoor scenarios and seven daily activities, reflecting the resource constraints typical of ad-hoc IoT deployments. Unlike existing CSI datasets that rely on high-power network cards, ESP-Fi HAR demonstrates the feasibility of a scalable and energy-efficient HAR solution using commodity IoT nodes. We further provide benchmark deep learning models to evaluate recognition accuracy, robustness, and deployment feasibility. Experimental results reveal the effects of signal propagation, environmental dynamics, and temporal heterogeneity on recognition performance, offering practical guidance for real-world system deployment. By integrating dataset construction, IoT system design, and resource-aware evaluation, ESP-Fi HAR establishes a foundation for future research on collaborative and scalable WiFi sensing in ad-hoc IoT networks. To promote further research in this field, we have made the dataset and codes publicly available at <https://github.com/AutoSmartGroup/ESP-Fi-HAR>.

## 1. Introduction

The proliferation of Internet of Things (IoT) technologies has intensified the demand for continuous and unobtrusive sensing capabilities in applications such as smart homes and health monitoring [1]. In ad-hoc network environments, where devices must self-organize under stringent resource constraints, achieving scalable and privacy-preserving human activity recognition (HAR) remains a significant challenge [2,3]. To address the limitations of resource-constrained sensor nodes, energy-efficient strategies such as adaptive energy management and energy harvesting have been proposed [4,5]. Traditional sensing modalities, including cameras [6,7] and wearable sensors [8], face limitations in deployment scalability, energy efficiency, and user privacy. WiFi channel state information (CSI) has emerged as a promising device-free sensing technology that can capture fine-grained environmental changes, enabling applications in indoor localization [9–11], human–computer interaction [12], and health monitoring [13]. Its non-intrusive nature and compatibility with commercial hardware make it attractive for pervasive sensing.

However, existing CSI-based HAR systems often rely on high-end network interface cards (NICs), such as the Intel 5300 [14–16] or Atheros-based solutions [17,18], which incur substantial hardware costs (exceeding \$40) and demand complex driver configurations. More critically, these platforms consume 2–3 W during active operation, rendering them unsuitable for large-scale, energy-sensitive ad-hoc IoT deployments. As such, they are not designed for scalable ad-hoc IoT deployment due to their high power consumption and complex configuration, contradicting the core requirements of low-cost, ad-hoc, and resource-efficient networks.

In this work, we argue that leveraging low-cost, ultra-low-power commodity IoT hardware represents a viable path towards scalable and practical ad-hoc sensing. The ESP32 series, particularly the ESP32-C3, offers a highly integrated system-on-chip solution with 2.4 GHz WiFi/Bluetooth connectivity, a 32-bit RISC-V processor, and hardware-accelerated cryptographic support—all within a compact QFN 5 × 5 mm package. With a unit cost of \$5–15 and peak power consumption of only 0.13 W, it achieves a 12× cost reduction and over 95% power savings compared to conventional NIC-based platforms.

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Coupled with the open-source ESP-CSI-Tool [19], the ESP32-C3 enables stable and efficient CSI acquisition without external RF components, making it ideal for resource-constrained IoT sensing.

Using this platform, we construct and present ESP-Fi HAR, a multi-scenario CSI dataset designed specifically for HAR in ad-hoc IoT environments. The dataset comprises CSI measurements from eight participants performing seven activities — walking, running, falling, jumping, squatting, arm waving, and turning — across four distinct indoor settings: a corridor, office, meeting room, and laboratory. All data were collected using a systematic ESP32-C3-based sensing setup and are released in standardized formats to facilitate reproducibility.

The main contributions of this work are summarized as follows:

(1) We design and implement a complete low-power IoT data acquisition system based on the ESP32-C3 and release the resulting ESP-Fi HAR dataset, comprising CSI samples collected in four real-world environments. This serves as a valuable and deployment-realistic benchmark for ad-hoc sensing networks.

(2) We demonstrate the feasibility of high-accuracy HAR using ultra-low-cost and ultra-low-power commercial hardware, addressing the critical bottlenecks of deployment cost and energy efficiency inherent in traditional systems. This opens new possibilities for large-scale IoT perceptual networks.

(3) This study evaluates multiple models on the four indoor scenarios of the constructed dataset, verifying their effectiveness across different environments. In addition, the key factors affecting HAR performance and the associated challenges are discussed, providing guidance for future system design.

We organize the remaining paper as follows. Section 2 describes the related work. Section 3 details the implementation of the CSI data acquisition system for the ESP-Fi HAR dataset. Section 4 outlines the features of this dataset. Section 5 presents the experimental setup, evaluation protocols, and results for activity recognition. Section 6 provides discussion and analysis of the experimental findings. Finally, the paper is summarized in Section 7.

## 2. Related work

In the past decade, several real-world datasets related to HAR, particularly those utilizing WiFi CSI for detection, have been introduced. These datasets span various application fields such as smart homes [20, 21], health monitoring [22–24], environmental sensing [25], and action detection [26,27]. They have provided valuable resources for advancing HAR technologies in diverse scenarios. Below are some representative datasets that have made significant contributions to their respective fields, while also having certain limitations.

### 2.1. UT-HAR dataset

The UT-HAR dataset [14], as the first publicly available CSI dataset for HAR, includes seven activity categories. The data was collected using the Intel 5300 NIC in the 2.4 GHz frequency band. Although this dataset has provided an important foundation for subsequent research, it also has some shortcomings. First, the dataset lacks clear activity segmentation labels. Second, since the data was continuously collected, using the sliding window method for data segmentation easily generates a large number of duplicate samples. These issues constrain the practical applicability of the dataset in real-world scenarios, yet its value as a pioneering benchmark in academic research remains irreplaceable.

### 2.2. Widar dataset

The Widar dataset [15], one of the largest WiFi sensing datasets, focuses on gesture recognition tasks. This dataset includes 22 types of gestures, and the data was collected using the Intel 5300 NIC and  $3 \times 3$  antenna configuration across multiple environments. To eliminate the interference of environmental variations on the data, Widar employs the Body-Coordinate Velocity Profile processing method, which successfully extracts gesture-related features from the original CSI data, significantly enhancing the data's robustness. However, Widar relies on expensive hardware devices (such as the Intel 5300 NIC and specialized antenna configurations), which limits its use in cost-sensitive application scenarios.

### 2.3. NTU-Fi dataset

The NTU-Fi dataset is used for action and gait recognition, leveraging the Atheros CSI tool and AR9271 WiFi NIC for data collection. It includes tasks for HAR [28] and human identification (Human ID) [29]. It offers high-frequency domain resolution with CSI from 114 sub-carriers, essential for analyzing subtle human movements like gait patterns. This makes it ideal for complex action analysis, including gait recognition and walking state detection, and supports applications such as gait biometrics, behavioral analysis, and elderly care.

However, the Atheros AR9271 NIC has driver compatibility issues, particularly with certain operating systems or devices, which can affect the dataset's stability and deployment, especially on Linux systems or edge devices. Performance fluctuations across hardware environments may also impact data collection consistency and model training results. Despite these challenges, NTU-Fi remains a valuable resource for WiFi-based human action and gait recognition research.

### 2.4. WiAR dataset

The WiAR dataset [16] addresses HAR through WiFi CSI, encompassing 16 daily activities including raising hands, bending, and walking. Collected in diverse indoor environments (offices, living rooms, corridors) using an Intel 5300 NIC with  $3 \times 3$  MIMO configuration, the dataset captures activity-signal relationships under realistic multipath propagation conditions. Ten participants performed each activity 30 times, yielding 4800 samples for robust pattern analysis. Beyond conventional action recognition, this benchmark specifically investigates the intrinsic correlation between CSI dynamics and human kinematics, providing insights into cross-scenario adaptation under interference-prone environments.

### 2.5. ESP32 dataset

The ESP32 Dataset [30] supports HAR via WiFi CSI, covering 10 daily activities: Jumping jack (A01), Squatting (A02), Hand swing (A03), Walking towards Rx (A04), Walking towards Tx (A05), Bouncing basketball (A06), Jogging in place (A07), Forward bend 90° (A08), Sitting down (A09), and Standing (A10). Data were collected in the Information Science Lab of Multimedia University (Melaka Campus,  $8.3 \text{ m} \times 5.6 \text{ m}$ ) across three indoor setups: two line-of-sight (LOS) and one non-line-of-sight (NLOS) scenario. LOS scenarios featured Tx–Rx distances of 7 m and 5.2 m, with activities performed at the midpoint, while the NLOS scenario included a wooden wall barrier.

The Tx was a D-Link AX3000 router and the Rx an ESP32 microcontroller, operating under IEEE 802.11n (Wi-Fi 4) on channel 11 with 40 MHz bandwidth. CSI and RSSI measurements were captured using the ESP-CSI Tool and stored as CSV files, each containing 52 subcarriers per packet. Eight volunteers (3 females, 5 males, aged 22–25) performed 20 trials per activity, yielding 1600 trials per setup. The dataset can be used to develop and evaluate WiFi CSI-based HAR models, particularly for low-cost, long-range through-wall applications.

**Table 1**  
Comparison of datasets for activity recognition.

| Dataset         | Platform         | Category number | Category names                                                                                                                                               | Overall price | Data size                                      |
|-----------------|------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|------------------------------------------------|
| UT-HAR          | Intel 5300 NIC   | 7               | Lie down, Fall, Walk, Pick up, Run, Sit down, Stand up                                                                                                       | \$40–100      | (3, 30, 250)<br>(antenna, subcarrier, packet)  |
| Widar           | Intel 5300 NIC   | 22              | Push & Pull, Sweep, Clap, Slide, 18 types of Draws                                                                                                           | \$40–100      | (22, 20, 20)<br>(time, x-velocity, y-velocity) |
| NTU-Fi HAR      | Atheros CSI Tool | 6               | Box, Circle, Clean, Fall, Run, Walk                                                                                                                          | \$70–120      | (3, 114, 500)<br>(antenna, subcarrier, packet) |
| NTU-Fi Human-ID | Atheros CSI Tool | 14              | Gaits of 14 Subjects                                                                                                                                         | \$70–120      | (3, 114, 500)<br>(antenna, subcarrier, packet) |
| WiAR            | Intel 5300 NIC   | 16              | 16 types of Activities and Gestures                                                                                                                          | \$40–100      | /                                              |
| ESP32 Dataset   | ESP32            | 10              | Jumping jack, Squatting, Hand swing, Walking towards Rx, Walking towards Tx, Bouncing basketball, Jogging in place, Forward bend 90°, Sitting down, Standing | \$5–15        | (1, 52, –)<br>(antenna, subcarrier, packet)    |
| Wallhack1.8k    | ESP32            | 3               | No presence, Walking, Walking + arm-waving                                                                                                                   | \$5–15        | (2, 128, 400)<br>(antenna, subcarrier, packet) |
| ESP-Fi HAR      | ESP32            | 7               | Run, Jump, Walk, Squat, Turn, Arm swing, Fall                                                                                                                | \$5–15        | (1, 52, 950)<br>(antenna, subcarrier, packet)  |

## 2.6. Wallhack1.8k dataset

The Wallhack1.8k Dataset [31] is a WiFi CSI-based HAR dataset tailored for long-range through-wall scenarios. It was collected using ESP32-S3 microcontrollers in both LOS (18 m hallway) and NLOS (five adjacent rooms separated by 25 cm-thick brick walls) environments. Two antenna configurations were employed: a directional biquad antenna (10–12 dBi gain) and a printed inverted-F antenna (PIFA) enhanced with a plane reflector, with the biquad offering superior RSSI and long-range stability. The dataset features three activity classes: No presence, Walking, and Walking + arm-waving. CSI packets, sampled at 100 Hz, were segmented into 400-packet (4-second) intervals and transformed into  $400 \times 52$  spectrograms (time  $\times$  subcarriers) after Hampel filtering for outlier removal. Each segment includes activity labels and room-specific presence labels (across five zones), with the 1806 samples split into an 8:1:1 train-validation-test ratio across four subsets (LOS/NLOS  $\times$  biquad/PIFA). This publicly available resource provides a compact, low-cost benchmark for assessing ESP32-based through-wall HAR, facilitating reproducible evaluations of resource-efficient, privacy-preserving activity detection algorithms in diverse settings.

Recent research in ad-hoc networks has explored energy-efficient and coordinated sensing systems for resource-constrained environments [32,33], demonstrating the importance of designing scalable, low-power, and collaborative sensing solutions. As shown in Table 1, although many existing datasets have achieved notable success in their specialized domains, they largely rely on sophisticated hardware configurations, which not only incur high deployment costs but also require complex maintenance procedures. These inherent limitations fundamentally hinder their adaptability in cost-sensitive IoT ecosystems and restrict their scalability across diverse application scenarios.

In contrast, the WiFi CSI dataset constructed in this study using ESP32-C3 devices offers a cost-effective alternative without compromising performance. By releasing this dataset, we not only address the gap in ESP32-based WiFi CSI data acquisition but also provide a valuable foundational resource for human activity recognition, fall detection, and related applications.

## 3. CSI data acquisition system implementation

### 3.1. Testing platform

The setup for the CSI signal collection experiments is shown in Fig. 1. This study established a signal sampling and CSI data logging platform based on the ESP32-C3 device, a high cost-performance measurement solution that supports WiFi signal transmission and CSI data collection, demonstrating encouraging application potential in the field of instrumentation. Using the ESP-CSI-Tool, we ensured the accuracy and integrity of the data collection process. To ensure the broad applicability and realism of the test scenarios, data collection was conducted in four representative environments: corridor, office, meeting room, and laboratory.

### 3.2. Hardware specifications

The experiments were conducted on a laptop with 16 GB RAM, an RTX 4070 GPU, and Windows OS. Two ESP32-C3 devices were used for WiFi signal transmission and CSI data collection. The transmitter optimized signal strength to avoid nonlinear distortion, while the receiver collected CSI data using the ESP-CSI-Tool at 100 Hz. Data was stored in CSV format to preserve integrity and enable consistent model training.

Detailed hardware specifications, development environment, hardware integration, and firmware deployment workflow are provided in Appendix for reference.

### 3.3. ESP-CSI-tool setup

The CSI data acquisition framework was established using VS Code with the Espressif IDF extension. The ESP32-C3 devices were connected via USB, with serial parameters configured to ensure stable communication. Firmware was deployed following a build-flash-validate workflow using the ESP-CSI-Tool, ensuring real-time CSI stream stability and reliable CSV-formatted data collection.

Full installation and configuration details are provided in Appendix.

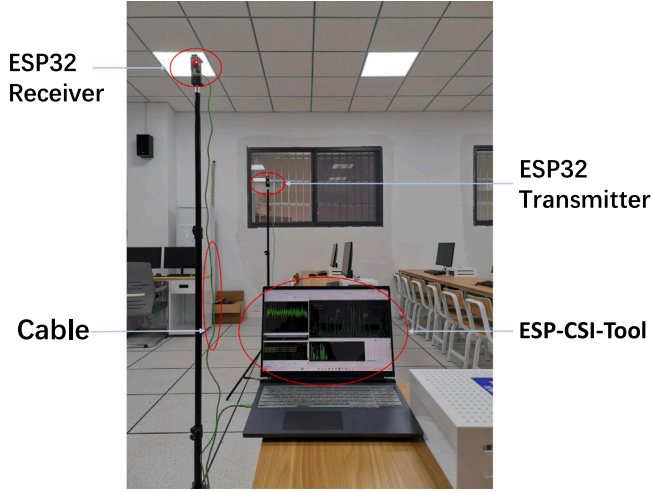


Fig. 1. Test bench for WiFi signal sampling.

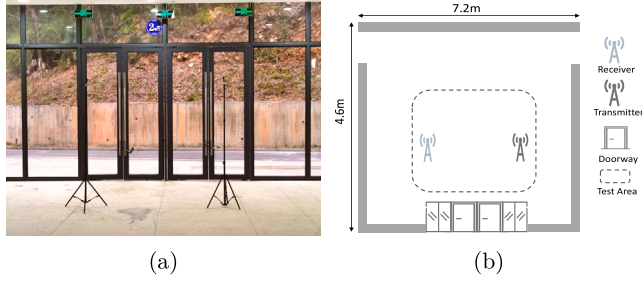


Fig. 2. Layout of corridor. (a) Scenario photo. (b) Training and test locations.

#### 4. ESP-FI HAR dataset overview

##### 4.1. Indoor field data collection environment

The data collection scenarios consisted of four typical indoor environments: corridor, office, meeting room, and laboratory. The collected data includes seven activities: walking (WK), running (RN), arm swinging (AS), squatting (ST), turning (TN), jumping (JP), and falling (FL), which are essential for assessing the model's robustness in real-world scenarios. The descriptions of these scenarios are as follows:

###### 4.1.1. Corridor scenario

As shown in Fig. 2, the experimental environment is a semi-open corridor located at No. 2 door on the first floor of the Industry-Education Integration Center. The outer side of the corridor consists of concrete and glass curtain walls, while the inner side is adjacent to the entrance of No. 2 door. Due to the presence of multiple entrances and high pedestrian traffic, this scenario exhibits a high level of environmental complexity characterized by dynamic interactions and interdependencies.

###### 4.1.2. Office scenario

As shown in Fig. 3, the experimental environment is a closed office space on the fourth floor of the Industry-Education Integration Center, furnished with desks, chairs, filing cabinets, and other standard office facilities. The layout is regular, with activities concentrated in fixed areas, and the relatively quiet environment contributes to its lower complexity compared to the corridor scenario.

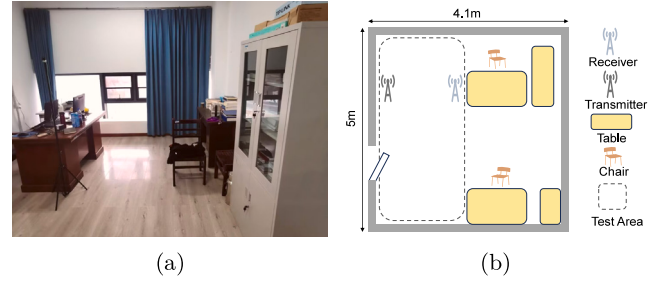


Fig. 3. Layout of office. (a) Scenario photo. (b) Training and test locations.

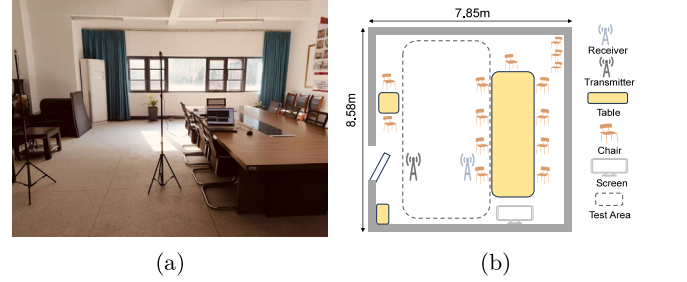


Fig. 4. Layout of meeting rooms. (a) Scenario photo. (b) Training and test locations.

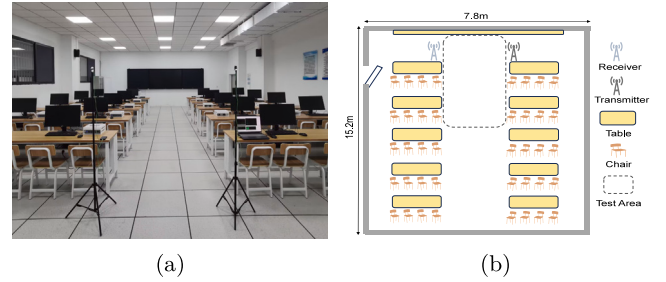


Fig. 5. Layout of laboratory. (a) Scenario photo. (b) Training and test locations.

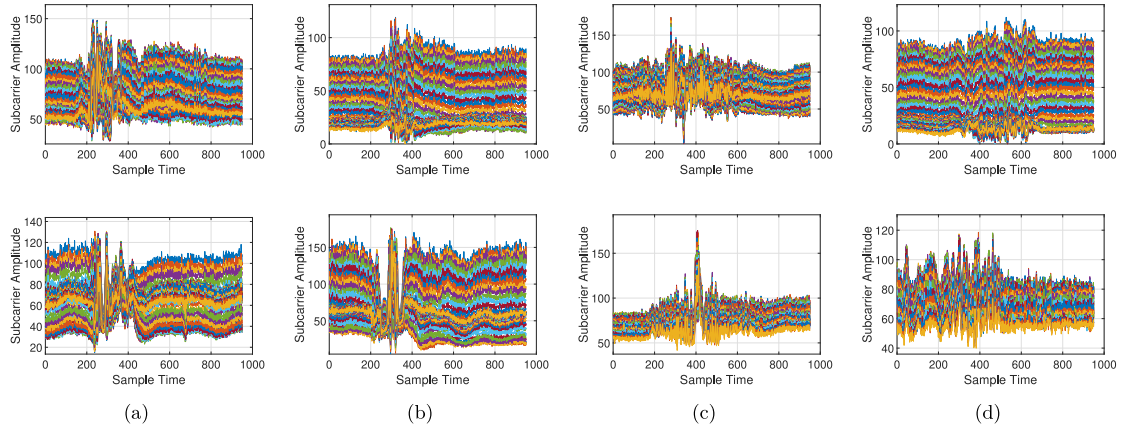
###### 4.1.3. Meeting room scenario

As shown in Fig. 4, the field data collection for this scenario was conducted in a meeting room on the fourth floor of the Industry-Education Integration Center. The experimental environment is a closed meeting room, equipped with a long table, chairs, projection equipment, and a whiteboard. The meeting room is relatively large, with an open layout, and most activities are concentrated around the conference table. The background noise is low, and the activities are relatively structured and static. Due to its static nature, the complexity of the meeting room scenario is moderate.

###### 4.1.4. Laboratory scenario

As shown in Fig. 5, the field data collection for this scenario was conducted in a laboratory located on the first floor of the Industry-Education Integration Center. The experimental setting is a standard laboratory environment, equipped with lab benches, various pieces of experimental equipment, storage cabinets, and other essential scientific research facilities. The simultaneous operation of multiple devices within the laboratory may generate certain interferences, and these factors collectively make the laboratory scenario one of the more complex among all the tested environments.

Fig. 6 illustrates the CSI amplitude plots captured by the ESP32 in a corridor scenario. In these plots, the x-axis represents a continuous sequence of data packets collected over approximately 10 s, while the



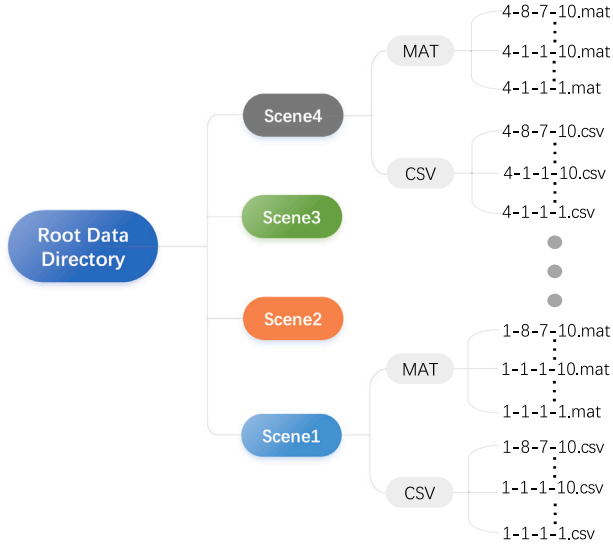
**Fig. 6.** CSI samples of four human activities of different people at ESP-Fi collected by ESP32. (a) Arm waving amplitude. (b) Jumping amplitude. (c) Walking amplitude. (d) Falling amplitude.

**Table 2**  
Physical characteristics of participants in ESP-Fi HAR.

| Participant ID | Height (cm) | Weight (kg) |
|----------------|-------------|-------------|
| 1              | 175         | 70          |
| 2              | 160         | 55          |
| 3              | 180         | 75          |
| 4              | 165         | 60          |
| 5              | 170         | 68          |
| 6              | 162         | 58          |
| 7              | 178         | 72          |
| 8              | 172         | 65          |

**Table 3**  
Activity categories and indoor scenarios.

| ID | Activity categories | ID | Indoor scenarios |
|----|---------------------|----|------------------|
| 1  | Run                 | 1  | Corridor         |
| 2  | Walk                |    |                  |
| 3  | Jump                | 2  | Office           |
| 4  | Squat               |    |                  |
| 5  | Arm Wave            | 3  | Meeting Room     |
| 6  | Turn                |    |                  |
| 7  | Fall                | 4  | Laboratory       |



**Fig. 7.** Dataset organization overview.

y-axis indicates the real-time amplitude variations of 52 subcarriers. Analysis of data from two participants — Participant 1 (upper plot) and Participant 2 (lower plot) — reveals that different activities produce distinct feature patterns. Moreover, the observed features vary across individuals. Table 2 details the physical attributes of the ESP-Fi HAR participants, which helps further elucidate the differences among individuals.

#### 4.2. Data storage format

All collected data is stored in CSV format, with each CSV file corresponding to a complete collection process of a specific action.

The duration of each collection is 10 s, containing approximately 1000 samples. To adapt to the training requirements of deep learning models, we use a Python script to convert CSV files into dual MAT format files, which store CSI amplitude and phase information respectively. Each MAT file adopts a data structure of  $1 \times 52 \times 950$ , corresponding to the number of antennas, the number of subcarriers, and the number of time sampling points, respectively.

Although the phase data files are completely retained, we still prioritize amplitude information as input features, mainly due to the essential differences in their characteristics: the raw phase of a single antenna has random offsets, leading to poor distribution stability; while the CSI amplitude has higher time-domain stability and is more suitable for WiFi sensing scenarios. In addition, basic noise reduction techniques (e.g., sliding-window filtering) can effectively suppress the high-frequency noise in the amplitude data, significantly improving its robustness to environmental interference while maintaining the integrity of core features.

As shown in Fig. 7, the dataset is hierarchically organized across three dimensions: scenario, participant, and activity. The dataset follows a structured naming convention: X-Y-Z-M, where:

- X denotes the scenario ID (ranging from 1 to 4, corresponding to four distinct indoor scenarios),
- Y indicates the participant ID (1 to 8, representing eight participants),
- Z specifies the activity category ID (1 to 7, mapped to seven predefined actions),
- M represents the collection trial number (1 to 10, indexing ten repeated trials per participant per scenario).

Table 3 further provides an explicit mapping between identifiers, indoor scenarios, and activity categories. This multi-level design ensures systematic data collection while facilitating efficient experimental data retrieval through a standardized naming system.

**Table 4**

Performance comparison of ML And deep learning models across four scenes.

| Model                          | Corridor     |             |              | Office       |             |              | Meeting room |             |              | Laboratory   |             |              |
|--------------------------------|--------------|-------------|--------------|--------------|-------------|--------------|--------------|-------------|--------------|--------------|-------------|--------------|
|                                | Acc (%)      | Acc Std (%) | F1 (%)       | Acc (%)      | Acc Std (%) | F1 (%)       | Acc (%)      | Acc Std (%) | F1 (%)       | Acc (%)      | Acc Std (%) | F1 (%)       |
| <b>Machine Learning Models</b> |              |             |              |              |             |              |              |             |              |              |             |              |
| SVM                            | 24.64        | 8.01        | 19.00        | 24.11        | 7.09        | 19.08        | 20.00        | 9.20        | 15.83        | 19.82        | 10.61       | 14.31        |
| RF                             | 23.57        | 7.14        | 19.01        | 21.25        | 6.16        | 17.19        | <b>24.64</b> | 12.37       | <b>21.82</b> | 17.68        | 7.59        | 13.47        |
| LR                             | 19.29        | 7.63        | 16.87        | 22.86        | 3.98        | 19.51        | 16.43        | 5.05        | 12.66        | 14.82        | 2.36        | 11.14        |
| <b>Deep Learning Models</b>    |              |             |              |              |             |              |              |             |              |              |             |              |
| LSTM                           | 32.15        | 13.40       | 29.82        | 38.02        | 14.21       | 33.80        | 36.20        | 12.04       | 33.84        | 32.73        | 13.25       | 30.21        |
| GRU                            | 30.93        | 3.78        | 28.74        | 38.02        | 8.63        | 33.00        | 34.55        | 4.21        | 31.17        | 28.38        | 8.98        | 24.56        |
| ResNet18                       | <b>63.59</b> | 14.91       | <b>60.81</b> | <b>70.11</b> | 19.60       | <b>64.00</b> | <b>65.53</b> | 18.22       | <b>61.26</b> | <b>58.33</b> | 22.73       | <b>49.14</b> |
| EfficientNet-Lite              | 61.60        | 10.60       | 57.13        | 63.74        | 8.74        | 58.80        | 61.98        | 5.14        | 57.93        | 50.35        | 6.96        | 46.27        |
| CNN                            | 57.55        | 11.43       | 53.20        | 48.31        | 8.02        | 43.80        | 48.26        | 11.14       | 42.70        | 52.61        | 12.19       | 46.89        |
| Transformer                    | 54.43        | 12.30       | 49.85        | 41.93        | 9.47        | 38.20        | 47.51        | 10.71       | 42.43        | 50.33        | 15.11       | 46.35        |
| MobileNetV3                    | 53.64        | 13.24       | 47.28        | 51.82        | 11.85       | 47.00        | 53.28        | 15.18       | 46.80        | 43.31        | 16.25       | 38.43        |

#### 4.3. Specific applications of the dataset

The ESP-Fi HAR dataset, collected in real-world environments, is designed to advance WiFi-based HAR under practical deployment constraints. By capturing CSI data in authentic scenarios, this dataset bridges the gap between controlled laboratory experiments and real-world implementations. Key applications include the following domains:

##### 4.3.1. Smart home automation

In smart home automation, the ESP-Fi HAR dataset supports robust human activity recognition in dynamic residential environments. Its multi-scenario coverage and adaptive mechanisms enable the system to accommodate environmental changes, such as furniture rearrangements. By leveraging the dual-core architecture of the ESP32-C3, one core continuously processes CSI data in real time, while the other monitors environmental variations, facilitating automatic spatial calibration through adaptive channel impulse response windowing.

##### 4.3.2. Health monitoring

In the health monitoring domain, this dataset exploits the stable spatiotemporal characteristics of CSI amplitude patterns to improve detection accuracy via adaptive multi-domain fusion. The ESP32-C3's RISC-V CPU, with its hardware-accelerated IIR filter, efficiently suppresses transient noise from moving individuals, enabling applications like Parkinsonian gait recognition by detecting gait abnormalities that may indicate neurological disorders. Additionally, automatic channel state tracking using the built-in inertial measurement unit compensates for sensor displacement, while context-aware fusion separates activity patterns from background dynamics for reliable cross-location comparisons.

##### 4.3.3. Elderly care

For elderly care, the dataset underpins reliable, non-invasive fall detection across diverse living settings through its phase-free data design and adaptive RF shielding, which reduces 2.4 GHz interference by 21 dB from neighboring WiFi networks. The energy-efficient ESP32-C3 nodes, operating under context-triggered wake-up protocols, support continuous 24/7 monitoring. This high cost-performance measurement solution highlights its promising potential for practical deployment in environments such as retirement communities and studio apartments.

#### 4.4. Dataset limitations

Although the ESP-Fi HAR dataset provides a comprehensive baseline for low-power IoT sensing, several limitations should be acknowledged. First, the participant pool is relatively compact, which may influence the model's generalization across diverse physiological profiles. Second, data collection was conducted in controlled indoor environments; thus,

the robustness of models trained on the dataset in highly dynamic outdoor or large-scale industrial scenarios remains to be validated.

Furthermore, the current dataset focuses on single-user activity and does not explicitly model multi-user interactions. While benchmarks such as WiMANS [34] have explored multi-user WiFi sensing, achieving reliable CSI annotation and signal decoupling under low-power, ad-hoc IoT constraints remains a significant technical hurdle due to hardware-level sampling limitations and the lack of inter-node synchronization. We have therefore prioritized establishing a rigorous and reproducible single-user benchmark for ESP32-class devices. Expanding the dataset to include multi-user interference modeling and collaborative cross-device sensing represents a primary direction for our future research.

## 5. Experiments and results

### 5.1. Baseline models and experimental setup

To systematically evaluate WiFi CSI-based HAR methods, we designed a set of baseline architectures, including both traditional machine learning (ML) models and modern deep learning (DL) models.

#### 5.1.1. Machine learning models

Three representative ML models were implemented: Support Vector Machine (SVM) [35], Random Forest (RF) [36], and Logistic Regression (LR) [37]. These models serve as classical references for comparison with more advanced DL architectures and help assess the ability of traditional methods to capture spatio-temporal features in various indoor scenes, including Corridor, Office, Meeting Room, and Laboratory.

#### 5.1.2. Deep learning models

For deep learning, we implemented both classical and modern architectures, including LSTM [38], GRU [39], CNN [40], ResNet18 [41], EfficientNet-Lite [42], Transformer [43], and MobileNetV3 [44], following configurations validated in previous HAR studies. All DL models were trained using the AdamW optimizer with consistent hyperparameter settings. All models were trained under identical data splits and optimization settings. ML models employed standard cross-validation procedures. Model performance was evaluated using two standard metrics: Accuracy (Acc) and F1-score, with the F1-score as the primary metric, calculated as:

$$F1 = 2 \times \frac{\left( \frac{TP}{TP+FP} \right) \times \left( \frac{TP}{TP+FN} \right)}{\frac{TP}{TP+FP} + \frac{TP}{TP+FN}} \quad (1)$$

where TP represents true positive predictions, FP false positives, and FN false negatives.

**Table 5**  
Model complexity metrics and standardized CPU inference latency.

| Model             | Parameters (M) | FLOPs (M) | Avg CPU latency (ms) | Std latency (ms) |
|-------------------|----------------|-----------|----------------------|------------------|
| SVM               | 22.38          | –         | 15.02                | 4.49             |
| RF                | 0.03           | –         | 7.18                 | 0.80             |
| LR                | 0.35           | –         | 0.09                 | 0.12             |
| ResNet18          | 11.21          | 609.72    | 12.31                | 1.26             |
| CNN               | 3.96           | 130.49    | 2.63                 | 3.75             |
| LSTM              | 0.26           | 13.55     | 1.17                 | 1.17             |
| GRU               | 0.42           | 21.61     | 1.66                 | 1.66             |
| Transformer       | 0.12           | 2.27      | 1.43                 | 0.32             |
| MobileNetV3       | 1.67           | 244.47    | 3.79                 | 1.59             |
| EfficientNet-Lite | 5.40           | 392.54    | 7.93                 | 2.31             |

## 5.2. Test results

To evaluate the effectiveness of the models, comprehensive experiments were conducted across four representative indoor scenes: Corridor, Office, Meeting Room, and Laboratory. All models were trained and evaluated under identical data splits and optimization settings using an 8-fold leave-one-subject-out (LOSO) cross-validation protocol to assess generalization performance.

The performance metrics include activity recognition accuracy, F1-score, model parameters, and computational complexity. Notably, we report average CPU inference latency (Avg CPU Latency) and its standard deviation (Std Latency) as a hardware-agnostic benchmark of model efficiency. This approach is specifically chosen to decouple model-level algorithmic efficiency from specific hardware bottlenecks. For artificial intelligence of things (AIoT) system-on-chip (SoC) devices like the ESP32-C3, which lack hardware floating-point units, executing FP32 operations via software emulation can increase latency by 6.5× to 60× compared to high-performance platforms [45]. Therefore, the standardized CPU-based results provide a more reproducible and intrinsic reference for the research community, reflecting the computational demands of different architectures without being confounded by MCU-specific hardware limitations.

The CPU latency evaluation follows common benchmarking practices [46]. Specifically, on a standard CPU platform, each model was set to evaluation mode with trained weights loaded. We used FP32 precision for forward passes of a single input sample (size  $[1 \times 1 \times 950 \times 52]$ ). Each evaluation was preceded by 200 warm-up iterations to eliminate initialization overhead, and inference times were recorded across multiple runs to calculate the mean, standard deviation, minimum, and maximum values.

### 5.2.1. Activity recognition accuracy

Accuracy is a classical evaluation metric in HAR. As shown in Table 4, deep learning models consistently outperform traditional ML models across all scenes. Specifically, ResNet18 attains the highest recognition accuracies among the deep learning models across all scenes, while within the ML models, SVM shows relatively better performance in certain scenarios. These results indicate that deep architectures, with higher capacity for spatio-temporal feature extraction, provide significant advantages in WiFi CSI-based HAR.

### 5.2.2. Analysis of model complexity and computational efficiency

Efficiency is a critical bottleneck in AIoT-based HAR systems. As shown in Table 5, deep learning models exhibit varying trade-offs between architectural complexity and computational speed. Specifically, while ResNet18 achieves the highest accuracy, it also incurs the largest computational overhead with 609.72 M FLOPs and an average CPU latency of 12.31 ms. In contrast, lightweight architectures such as MobileNetV3 and Transformer demonstrate superior efficiency; the Transformer model, in particular, maintains a remarkably low footprint of 0.12 M parameters and 1.43 ms latency. Notably, although traditional ML models like SVM have fewer operations, their latency

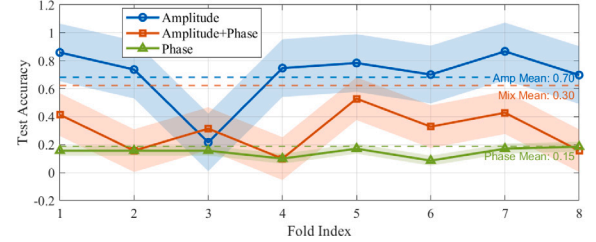


Fig. 8. Test accuracy for three input modalities.

(e.g., 15.02 ms) can exceed that of modern deep networks due to less optimized implementation for large-scale CSI data.

These results indicate that the benchmark DL architectures maintain a reasonable level of recognition capability while preserving sufficient algorithmic efficiency to meet the real-time requirements of edge-to-cloud sensing scenarios. It should be noted that these measurements are obtained on a standardized CPU platform, and their primary purpose is to provide a reproducible reference for comparing model performance and to guide future optimizations aimed at practical deployment.

### 5.2.3. F1-score

To assess the balance between precision and recall, we computed the macro F1-score. As detailed in Table 4, ResNet18 consistently achieves the highest F1-scores among DL models, underscoring its robustness across diverse indoor scenarios. In contrast, traditional ML models exhibit markedly lower F1-scores, revealing their constraints in capturing the intricate spatio-temporal features inherent in WiFi CSI data. Although ML models can discern basic patterns, DL approaches surpass them in delivering superior accuracy and balanced class recognition.

## 6. Comments and discussion

In this section, unless otherwise noted, all analyses are based on data collected in the Office scenario. Focusing on this representative environment allows us to examine model performance and feature behavior under realistic conditions. All reported results use the best-performing backbone, ResNet18, to ensure consistent comparisons. Experiments follow a LOSO cross-validation protocol, resulting in an eight-fold evaluation corresponding to the eight participants.

### 6.1. Recognition accuracy by input modality

Fig. 8 presents the overall test accuracy for the three input modalities — Amplitude, Phase, and Amplitude+Phase — providing a summary comparison before the detailed analyses in the following subsections.

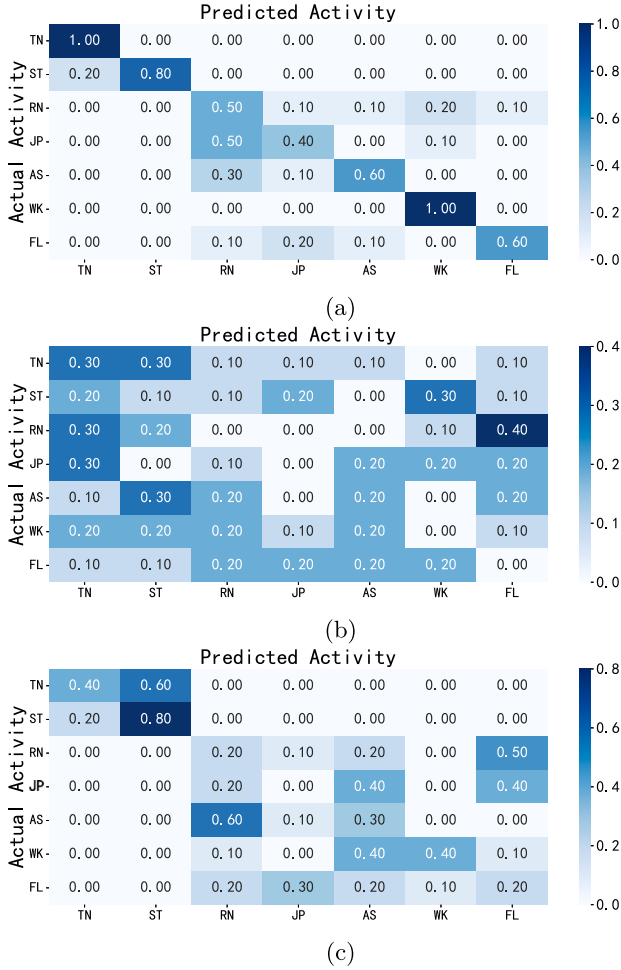


Fig. 9. Confusion matrices of Fold-6 under different input modalities using the ResNet18 model: (a) Amplitude, (b) Phase, and (c) Amplitude-Phase Mixed Input.

#### 6.1.1. Amplitude-only modality

As shown in Fig. 9(a), the amplitude-only model achieves the highest recognition accuracy across nearly all activity categories. The confusion matrix shows relatively strong diagonal dominance, indicating that amplitude features provide good discriminative capability for human activity recognition in the office environment.

#### 6.1.2. Phase-only modality

The phase-only results in Fig. 9(b) show noticeably lower accuracy, with widespread misclassifications across several activities. This confirms that raw phase measurements, affected by random hardware-induced phase offsets and multipath distortions, fail to provide clean discriminative features without preprocessing.

#### 6.1.3. Amplitude-phase fusion modality

As shown in Fig. 9(c), the recognition performance with amplitude-phase fusion input is lower than that of amplitude alone. This indicates that, without proper phase preprocessing, incorporating raw phase information does not provide complementary benefits and may slightly degrade performance. Advanced preprocessing such as phase unwrapping or offset correction may be necessary to fully exploit any complementary information.

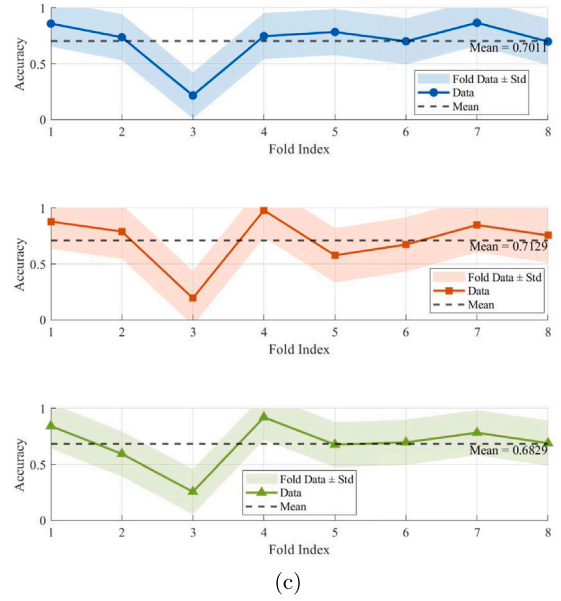


Fig. 10. Recognition accuracy for different amplitude preprocessing methods. (a) Unprocessed. (b) Moving average denoising. (c) Wavelet denoising.

### 6.2. Recognition accuracy by preprocessing methods

To evaluate the impact of amplitude preprocessing on HAR, we compared no preprocessing, moving average denoising, and wavelet denoising inputs. Fig. 10 shows that the different preprocessing methods result in some variation in recognition performance, reflecting the impact of noise reduction on model accuracy.

#### 6.2.1. No preprocessing

As shown in Fig. 10(a), the model trained on raw amplitude achieves moderate recognition accuracy across folds. While some activities are classified correctly, the fold-to-fold variability is relatively high, reflecting the impact of measurement noise and temporal fluctuations in the CSI amplitude. This indicates that raw amplitude contains useful motion information but is sensitive to noise, which can reduce consistency in recognition performance.

#### 6.2.2. Moving average denoising

Fig. 10(b) illustrates the results after applying a moving average filter to the amplitude input. The smoothing effect reduces short-term fluctuations and improves recognition accuracy for most folds. However, some extreme values are attenuated, which may slightly decrease the discriminative ability for activities with abrupt motions. Overall, moving average denoising provides a simple and effective way to enhance stability in the amplitude features.

#### 6.2.3. Wavelet denoising

As shown in Fig. 10(c), wavelet-based denoising aims to suppress high-frequency noise while preserving motion-related variations. However, the average recognition accuracy across folds is slightly lower than that of the raw amplitude input. While wavelet filtering reduces noise, it may also remove some useful signal information, highlighting the trade-off between noise suppression and signal preservation in amplitude-based HAR.

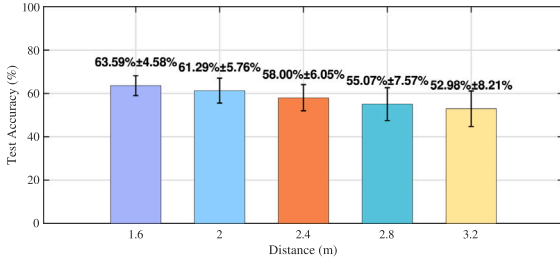


Fig. 11. Recognition accuracy at different transmitter–receiver distances.

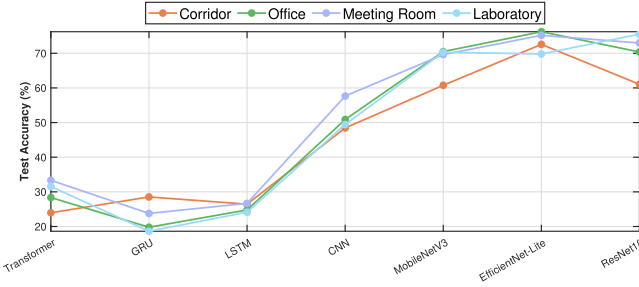


Fig. 12. Cross-scene accuracy comparison of different models across four scenes.

### 6.3. Recognition accuracy under distance variation

To evaluate the impact of distance on activity recognition performance, we analyzed how WiFi signal strength and recognition accuracy vary with increasing distance between the data collection device and the target. The variation in recognition accuracy with distance is illustrated in Fig. 11, which shows a clear downward trend as distance increases. Our results demonstrate that signal attenuation becomes more pronounced at larger distances, leading to reduced classification accuracy. Specifically, the degradation in signal integrity at extended ranges introduces noise and reduces feature discriminability, directly impacting the model's ability to recognize activities. This highlights the need for further optimization of signal processing techniques and data augmentation strategies to enhance the model's robustness in large or complex environments.

### 6.4. Cross-scene accuracy evaluation

We conducted cross-scene evaluations to assess model robustness. Each model was trained on data from three of the four representative scenes and tested on the remaining scene. The results are summarized in Fig. 12.

In the corridor, EfficientNet-Lite achieved the highest accuracy (72.57%), followed by ResNet18 (61.05%) and MobileNetV3 (60.76%), while Transformer performed the worst (23.96%). In the office, EfficientNet-Lite again led (76.22%), with ResNet18 and MobileNetV3 performing similarly around 70%, and GRU yielding the lowest accuracy (19.79%).

For the meeting room, EfficientNet-Lite and ResNet18 achieved comparable performance (75.17% and 72.97%, respectively), whereas Transformer was the lowest (33.33%). In the laboratory, ResNet18 reached the highest accuracy (75.52%), followed by MobileNetV3 (70.31%) and EfficientNet-Lite (69.79%), with Transformer again performing worst (31.54%).

Overall, CNN-based models, particularly ResNet18 and lightweight CNNs such as EfficientNet-Lite, maintain relatively stable performance across diverse scenes, while recurrent and transformer-based models are more sensitive to environmental variations.

Table 6

Duration statistics of seven activities in the office scenario (in seconds).

| Class | Mean   | Median | Std    | Min  | Max  | IQR  |
|-------|--------|--------|--------|------|------|------|
| TN    | 2.7453 | 1.940  | 2.5656 | 0.02 | 9.49 | 3.72 |
| ST    | 3.4216 | 2.885  | 2.6027 | 0.01 | 8.18 | 4.72 |
| RN    | 2.9051 | 1.660  | 2.7583 | 0.01 | 8.21 | 5.43 |
| JP    | 2.5256 | 0.760  | 2.8441 | 0.02 | 7.59 | 5.68 |
| AS    | 1.6270 | 0.850  | 2.1808 | 0.01 | 7.59 | 1.46 |
| WK    | 2.9342 | 1.605  | 2.7534 | 0.02 | 8.50 | 4.99 |
| FL    | 3.4372 | 2.315  | 2.9023 | 0.01 | 8.06 | 5.84 |

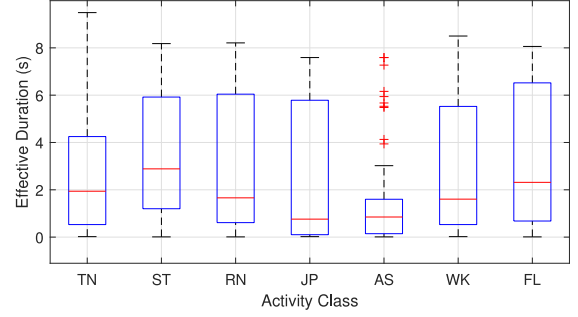


Fig. 13. Boxplot of effective action duration distributions across the seven activity classes in the office scenario.

### 6.5. Influence of duration imbalance on model decision behavior

To investigate the potential impact of activity duration imbalance on model performance, we analyzed the effective action duration distributions of the seven activity classes in the office scenario. As shown in Fig. 13, significant inter-class differences are observed. As summarized in Table 6, dynamic activities such as RN and JP exhibit higher dispersion (Std/IQR), whereas static or short-duration actions like AS and TN are more concentrated. This indicates that different activities possess inherent temporal discriminability, and the model may partially exploit duration as a discriminative cue.

However, duration does not appear to be the dominant factor in model decisions. Firstly, classes with similar mean durations, such as ST and FL, are well distinguished in both representative folds of the 8-fold cross-validation shown in Figs. 14 and 10(a), with minimal confusion (e.g., ST → TN 20%; FL → RN/JP/AS 40%). This suggests that the model relies more on spatiotemporal CSI features rather than simple duration cues. Secondly, activities with significantly different durations, such as RN and JP, still exhibit multi-directional confusion (RN → WK/FL/AS; JP → RN/WK), indicating that dynamic CSI patterns rather than temporal length primarily drive misclassification. Finally, the short-duration AS shows confusion across multiple longer-duration classes (RN, JP, WK), further confirming that decisions are mainly guided by CSI amplitude or temporal features, rather than raw duration alone.

Consistent patterns across both confusion matrices reveal that activities with higher duration dispersion (RN, JP) are more prone to misclassification, whereas classes with similar duration but distinct motion patterns (ST, FL) are almost perfectly separated. These findings indicate that the model's decision space is not dominated by activity duration, but robustly relies on richer CSI-based features.

### 6.6. Implications for future work

Based on the above analysis, this paper outlines several future research directions, aiming to further advance WiFi CSI-based human activity recognition models from three complementary perspectives, namely input modality design, cross-environment robustness enhancement, and system-level scalability, so as to address the limitations of existing approaches in complex real-world scenarios.

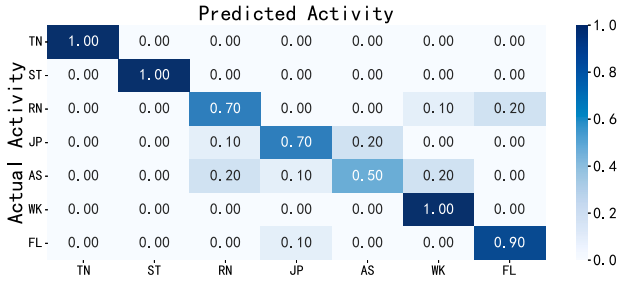


Fig. 14. Confusion matrix of seven activities in the office (one fold of 8-fold cross-validation).

First, with respect to input types and preprocessing strategies, future work may explore dynamic fusion mechanisms of amplitude and phase information, together with more advanced preprocessing techniques. Although existing studies indicate that amplitude-based inputs generally achieve superior recognition accuracy, selectively incorporating phase information through adaptive weighting schemes may help suppress noise in scenarios where amplitude patterns are highly similar, thereby improving discriminative capability.

Second, to enhance cross-scene robustness under diverse deployment environments, domain adaptation techniques can be further investigated to mitigate domain shifts caused by environmental variations, enabling stable model generalization across heterogeneous scenarios. One promising direction is the adoption of lightweight supervised calibration strategies that operate with only a small number of labeled samples in the target domain. Another direction involves joint modeling of spatial and temporal features, which enables feature-level cross-scene adaptation without requiring target-domain annotations. Future studies may further optimize these approaches and explore more efficient and generalizable domain adaptation frameworks.

Furthermore, from the perspective of system scalability, although this work establishes a rigorous single-user benchmark, extending the dataset and models to multi-user activity recognition remains a critical next step. Addressing signal disentanglement and reliable annotation on low-power devices is essential for supporting complex and realistic social sensing scenarios.

Moreover, multi-transmitter and multi-receiver CSI fusion represents a promising avenue for improving sensing reliability and system scalability [47,48]. By leveraging the spatial diversity offered by multi-link configurations, these frameworks can provide higher-dimensional data representations, which are instrumental for HAR in diverse settings. The transition from single-link observations to multi-link collaborative sensing potentially mitigates the limitations of individual link instability, offering a feasible roadmap for scaling benchmarks towards more complex and heterogeneous environments.

Meanwhile, to alleviate the bias introduced by action duration imbalance in model decision-making, future work may incorporate temporal normalization mechanisms, such as duration-aware resampling strategies or standardized event segmentation. These approaches can reduce the model's dependence on long-duration activities, promote fairer predictions, and improve robustness to temporal heterogeneity.

Finally, at the system deployment level, future research may further explore Edge-to-Cloud architectures, in which ESP32-series devices serve as low-power front-end units for CSI data acquisition and lightweight preprocessing, while computationally intensive inference tasks are offloaded to edge servers or the cloud. This architecture strikes a balance between real-time responsiveness and on-device computational constraints, providing an efficient and scalable WiFi-based human activity recognition solution for resource-constrained AIoT systems.

## 7. Conclusion

This paper presents ESP-Fi HAR, an open-access WiFi CSI dataset collected via a scalable and ultra-low-cost sensing system based on ESP32-C3 nodes. Experiments demonstrate that commercial off-the-shelf IoT hardware can serve as a viable alternative to traditional high-cost NICs (e.g., Intel 5300), achieving approximately a tenfold reduction in cost and over 95% reduction in power consumption. The dataset covers four distinct environments, and combined with the lightweight sensing node design, it effectively bridges the gap between controlled laboratory validation and deployment under resource-constrained conditions. Through systematic evaluation of various deep learning and lightweight models (e.g., MobileNetV3, EfficientNet-Lite, CNN), this study validates the feasibility and robustness of the low-cost ESP32 platform for HAR. In summary, this work provides a practical solution for building large-scale, low-cost, and resource-efficient WiFi sensing networks, and lays the foundation for future applications in distributed, multi-environment, and privacy-preserving scenarios.

## CRediT authorship contribution statement

**Zhiwei Wen:** Writing – original draft, Validation, Software, Methodology, Data curation. **Yanlin Ruan:** Writing – review & editing, Supervision, Methodology, Formal analysis, Conceptualization. **Xiaoye Wang:** Validation, Software, Resources, Investigation. **Junjie Zhou:** Writing – original draft, Validation, Software, Data curation. **Hongliang Gao:** Supervision, Project administration, Funding acquisition, Conceptualization. **Tao Li:** Visualization, Supervision, Funding acquisition, Formal analysis.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A. ESP32-C3 detailed specifications

The ESP32-C3 is a highly integrated microcontroller designed for wireless communication and IoT applications, and its functional architecture is illustrated in Fig. 15. Detailed hardware specifications used in our experiments are as follows:

- **CPU:** RISC-V single-core, 160 MHz
- **Memory:** 400 KB SRAM, 384 KB ROM, 4 MB Flash
- **Wireless:** WiFi 4 (802.11 b/g/n) and Bluetooth Low Energy (BLE)
- **GPIO and Interfaces:** 22 GPIO pins, 3 SPI interfaces, 1 UART interface, 1 I2C interface
- **Power Consumption:** 5  $\mu$ A in deep sleep mode
- **Integration:** On-chip RF components including integrated antenna switch, RF balancer, power amplifier, low-noise receiver amplifier, and power management modules

This configuration allows the ESP32-C3 to perform stable CSI data collection with minimal external components, balancing performance and energy efficiency.

## Appendix B. ESP-CSI-tool installation and configuration

The CSI data acquisition framework was implemented using the ESP-CSI-Tool for ESP32-C3 devices, and its capture interface is illustrated in Fig. 16. The detailed setup is as follows:

### B.1. Development environment

- Visual Studio Code with Espressif IDF extension
- ESP-IDF framework, compiler toolchain, and device-specific libraries installed

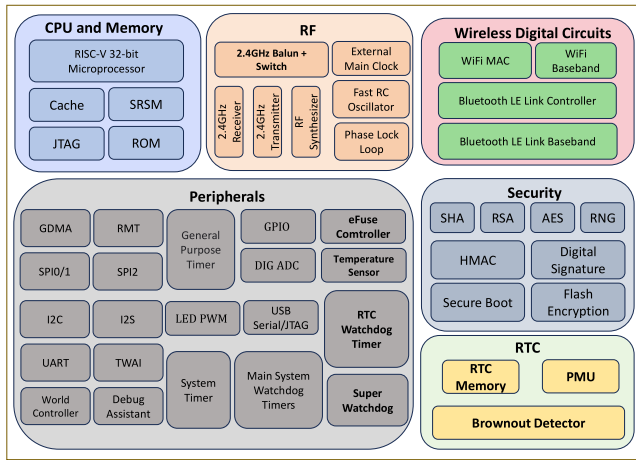


Fig. 15. ESP32 function block diagram.

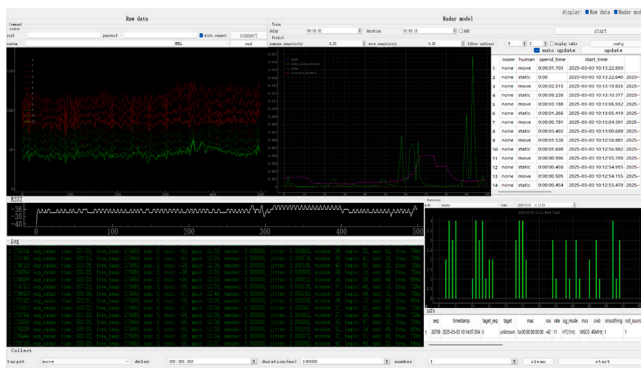


Fig. 16. Capture interface for ESP-CSI-Tool.

## B.2. Hardware integration

- ESP32-C3 devices connected via USB
- Serial communication configured with COM port (e.g., COM5) and baud rate 115200 bps
- Ensured stable communication for reliable CSI data transfer

## B.3. Firmware deployment workflow

1. **Build:** Compile firmware using ESP-IDF build system
2. **Flash:** Program transmitter and receiver with `csi-send` and `console-test` firmware
3. **Validate:** Monitor real-time CSI amplitude and phase streams using `esp-csi-tool.py` to ensure data integrity

All CSI data was recorded in CSV format for subsequent model training.

## Data availability

Data will be made available on request.

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